AUTOMATED SMART GATE PASS AND IDENTITY VERIFICATION SYSTEM FOR SECURE CAMPUS ACCESS

ABSTRACT

The Smart Gate Pass System is a digital solution designed to simplify and automate the process of managing gate passes in educational institutions and organizations. Traditional gate pass systems generally depend on manual registers, paper-based approvals, and physical verification, which can lead to delays, data loss, unauthorized entry, and difficulties in maintaining accurate records. The proposed system addresses these limitations by providing a centralized platform for creating, approving, monitoring, and managing gate pass requests.

The system allows users to submit gate pass requests by providing the required details, while authorized personnel can review the requests and approve or reject them based on the provided information. The system maintains gate pass records in a database, enabling easy retrieval and verification whenever required. Authentication and role-based access help ensure that only authorized users can perform administrative and approval-related activities. The system also provides a structured workflow that improves communication between students/users, administrators, and security personnel.

By replacing manual procedures with a computerized approach, the Smart Gate Pass System improves efficiency, transparency, security, and record management. It reduces paperwork and minimizes human errors while providing faster processing and better tracking of gate pass activities. Overall, the system offers a practical, user-friendly, and scalable approach for managing institutional gate pass operations.

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1. INTRODUCTION

In educational institutions and organizations, managing the exit of students or authorized users from the premises requires a proper approval process. Traditionally, exit requests may be handled manually through written permissions, registers, or direct communication with the concerned authority. Such methods can be time-consuming and may make it difficult to maintain and track records of approved and rejected requests. A systematic approach is therefore required to make the exit gate pass process simpler and more organized.

The Smart Gate Pass System is developed to provide a digital platform for requesting and managing exit gate passes. A user can submit a gate pass request by providing the necessary details, such as the reason for leaving and other required information. The request is then forwarded to the concerned authority for verification and approval. Based on the decision, the gate pass request can be approved or rejected, allowing the exit process to be carried out in a controlled manner.

The system maintains the details of submitted gate pass requests and their approval status, which helps in keeping proper records and reducing dependency on manual documentation. It also makes the request and approval process more convenient for both users and authorities.

1.1 Problem Statement

Traditional gate-pass practices can depend on handwritten forms, manual signatures, and visual verification. Such processes may work for a small number of requests, but they become harder to manage when requests increase or staff need to retrieve an older record. A paper record may show that permission existed, but it does not conveniently connect submission, parent confirmation, approval, identity verification, and actual exit.

The project therefore focuses on a unified system that connects request submission, approval, identity verification, and exit recording.

- Reduce dependence on disconnected manual records.
- Make approval status visible to the appropriate role.
- Provide an identity check at the point of campus exit.

- Maintain an audit trail of important actions.

1.2 Objectives

The main objective is to develop a web-based Smart Gate Pass System that makes campus exit processing organized, traceable, and easier to verify.

- Provide role-based access for students, HOD/department users, security guards, and administrators.
- Allow students to submit gate-pass requests with date, time, and reason.
- Record parent notification and confirmation.
- Allow the HOD to approve or reject requests and record remarks.
- Provide biometric verification at the security stage.
- Record verified exit events and retain an audit history.

1.3 Scope of the Project

The scope is limited to campus gate-pass management and identity verification. The supplied implementation supports student registration, authentication, request creation, parent confirmation, departmental review, security verification, administrator functions, and audit logging.

The project does not attempt to provide general campus navigation, GPS tracking, attendance automation, or a complete college ERP. Although the source folder is named smart GPS, the implemented application behavior is centered on the Smart Gate Pass workflow rather than geographic location tracking.

1.4 Significance of the Project

The main value of the project is the connection between permission and identity. A gate pass is useful only when it is connected to a particular student and the security process can verify that identity. The application therefore combines administrative approval with a biometric check at the point of exit.

For students, the process becomes easier to follow. For the department, requests can be reviewed in one place. For security personnel, approved requests and identity verification are available together. For administrators, student records and audit events provide a broader view of activity.

2. LITERATURE SURVEY

S. No	Title of the Paper	Author of the Paper	Methodology they Used	Drawbacks of their Methodology	Research Gap/Gap Identified
1	Face Recognition Based Gate Pass System	Dr. Sunil Bhutada, Dr. Sreenivas Mekala, Mayukhi Gandham, Rishika Bhat, Ruchitha Upadhyayula	A web-based gate-pass system is developed using facial recognition. It manages gate passes and maintains electronic entry and exit records, reducing the use of manual paperwork.	The system focuses mainly on facial recognition and gate-pass management. A complete hierarchical approval process, such as parent confirmation followed by authorized approval, is not clearly addressed.	A system is required that connects student gate-pass requests, parent confirmation, authorized approval and facial verification.
2	Integrated RFID Solution for Student Entry and Exit Monitoring System	S. Norwahidayah Wahab, A. Luqman Bin Abdul Latif, M. Shah Shafie	The system uses RFID technology for automatic student entry and exit monitoring. RFID readers, Raspberry Pi, a servo motor, PHP and MySQL are used to record student movement.	RFID requires the student to carry and use an RFID card or tag. The system mainly monitors entry and exit and does not provide facial verification or a complete gate-pass approval process.	There is a need for an integrated system that provides digital gate-pass authorization, identity verification, approval management and automatic entry/exit records.

Table No.1 – Literature Survey

2.1 Research Gap

Existing systems use facial recognition, RFID, or digital gate passes separately. They do not provide a complete approval and verification process. The proposed Smart Gate Pass System addresses this gap by integrating the complete process of student registration, gate-pass request, parent confirmation, authorized approval, facial verification at the gate, and entry/exit record maintenance.

3. DESIGN

3.1 SYSTEM ARCHITECTURE

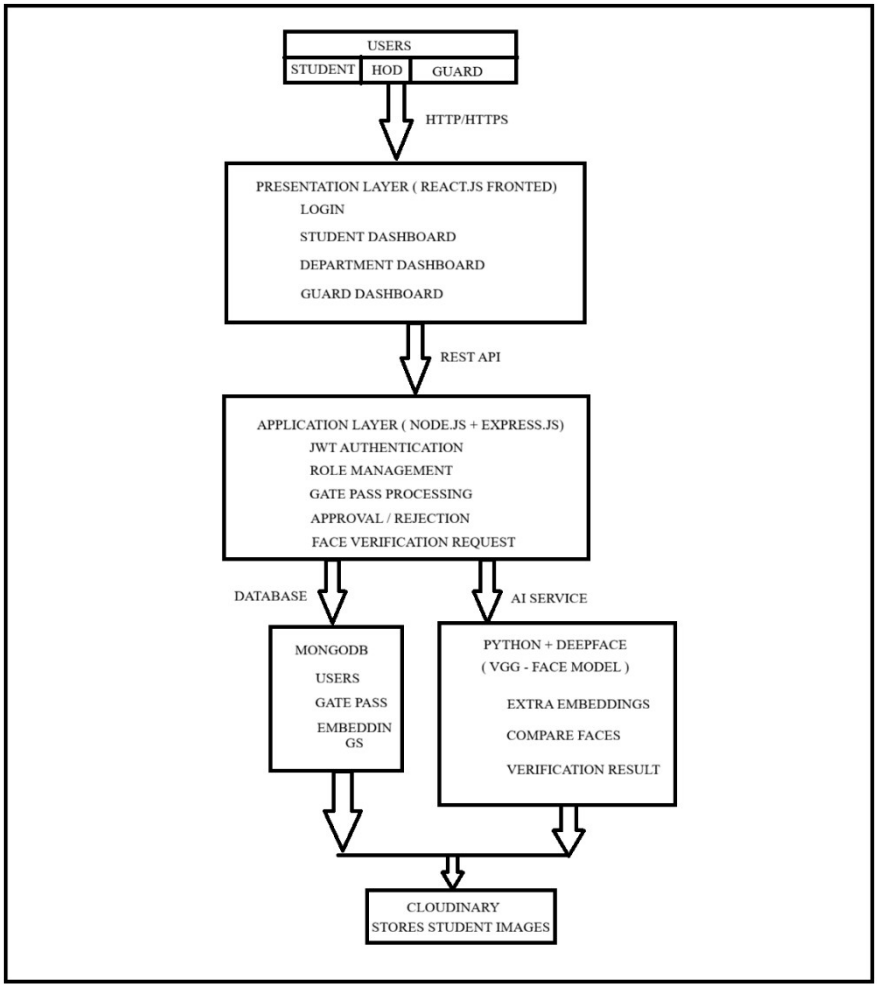


Fig 3.1 - System Architecture Flow Chart

The proposed Next-Gen Smart Pass System for Campus Security follows a web-based client-server architecture that integrates user management, gate pass processing, database storage, cloud image management, and facial recognition-based authentication. The architecture is designed to automate the complete gate pass workflow, including request submission, approval, identity verification, and digital record maintenance. The system consists of three major layers: the Presentation Layer, Application Layer, and Data Processing Layer. These layers work together to provide a secure, efficient, and scalable campus access management solution.

3.1.1 Presentation Layer

The Presentation Layer represents the user interface through which different users interact with the system. It provides separate portals based on user roles: students, department heads (HODs), and security guards.

- **Student Portal**

  The Student Portal allows students to access the system and perform gate pass-related activities. Students can register, log in, submit gate pass requests by providing necessary details such as date, time, and reason for leaving the campus, and track the approval status of their requests. During registration, student facial images are collected and uploaded for future identity verification. Students can also view their previous gate pass history and approval records.

- **HOD Portal**

  The HOD Portal enables department heads to manage student gate pass requests. HODs can view pending requests, review student details, analyze request history, and approve or reject applications based on the provided information. The approval decision is updated in the system database and is reflected in the security guard portal.

- **Security Guard Portal**

  The Security Guard Portal is used at the campus gate for final verification. Security guards can view approved gate pass requests and perform real-time face scanning of students. The system verifies the student's identity using facial recognition before allowing exit. Successful verification results in permission approval, while failed verification prevents unauthorized exit.

3.1.2 Application Layer (Node.js Backend Server)

The Application Layer acts as the core processing unit of the system. It is developed using Node.js and manages communication between the frontend interfaces, database, cloud storage, and facial recognition module. The backend performs the following operations:

- Handles user authentication and authorization.
- Maintains role-based access control.
- Processes student gate pass requests.
- Sends requests to respective HODs.
- Updates approval or rejection status.

- Retrieves approved requests for security verification.
- Communicates with MongoDB for data storage.
- Integrates facial recognition services.
- Maintains digital entry and exit records.

The Node.js server ensures smooth data flow between all system components and provides secure communication between different modules.

- **Database Layer (MongoDB)**

  MongoDB is used as the primary database for storing and managing system information. It provides flexible and scalable storage for user details, gate pass information, and verification records. The database maintains:

  - Student profiles
  - HOD details
  - Security guard accounts
  - Login credentials
  - Gate pass requests
  - Approval status
  - Facial recognition data references
  - Entry and exit logs

  MongoDB enables quick data retrieval and efficient management of large amounts of campus security data.

- **Cloud Storage Layer (Cloudinary)**

  Cloudinary is integrated into the system for secure storage and management of student facial images. Instead of storing images directly in the database, images are uploaded to Cloudinary, and their URLs are stored in MongoDB. The process includes:

  - Students upload a facial image during registration.
  - Image is securely stored in Cloudinary.
  - Cloudinary generates a unique image URL.
  - The URL is stored in MongoDB.
  - The image is retrieved during face verification.

  This approach reduces database storage requirements and improves image accessibility and scalability.

3.1.3 Facial Recognition Layer

The Facial Recognition Layer provides biometric authentication for verifying student identity. It uses the VGG-Face deep learning model to extract unique facial characteristics from images. The recognition process involves:

- Capturing the student's live facial image at the campus gate.
- Processing the image using the VGG-Face model.
- Extracting facial embeddings.
- Comparing live facial embeddings with stored facial embeddings.
- Calculating similarity using the Cosine Similarity algorithm.
- Generating the verification result.

If the similarity score meets the predefined threshold and the gate pass is approved, the student is allowed to leave the campus.

3.1.4 Face Verification Module

The Face Verification Module acts as the final security checkpoint of the system. It ensures that the person presenting the approved gate pass is the actual authorized student. The module performs the following functions:

- Captures real-time facial images.
- Extracts facial features.
- Compares facial embeddings.
- Confirms student identity.
- Generates access decisions.
- Creates digital exit logs.

This module prevents impersonation, fake gate passes, and unauthorized campus exits by combining biometric authentication with approval-based access control.

3.2 UML DIAGRAMS

3.2.1 Use Case Diagram

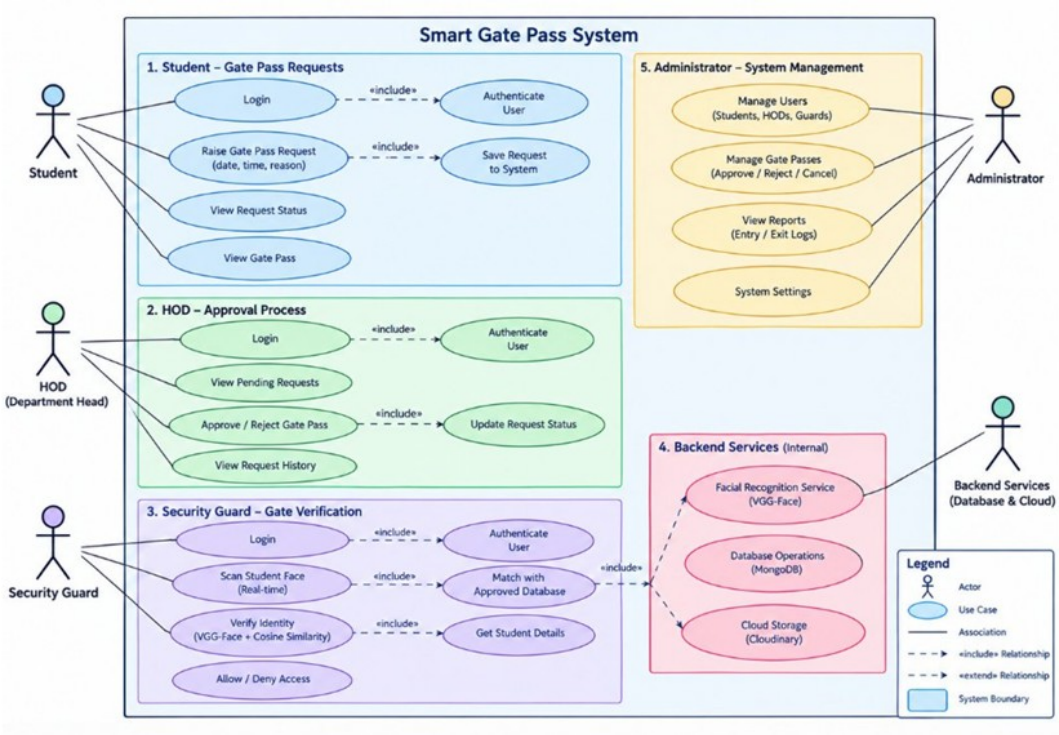


Fig 3.2.1 - Use Case Diagram

The use case diagram provides a high-level view of the interactions between the users (actors) and the functionalities (use cases) of the Next-Gen Smart Pass System. The system supports three primary user roles Student, Department (HOD), and Guard each with specific capabilities relevant to their responsibilities in the gate pass process.

3.2.2 Class Diagram

The class diagram represents the core structure of the Next-Gen Smart Pass for Campus Security. It defines a common User superclass extended by Student, Department, and Guard, each having login and logout functionality. Students can send and manage gate pass requests (Gate Pass Request), which include details like date, time, and reason. Departments can view, approve, or reject requests, while Guards verify approved students at the gate using the Face Recognition Service. This service handles image embedding and comparison for face authentication. The diagram clearly shows how each user type interacts with the system to ensure secure and efficient gate management.

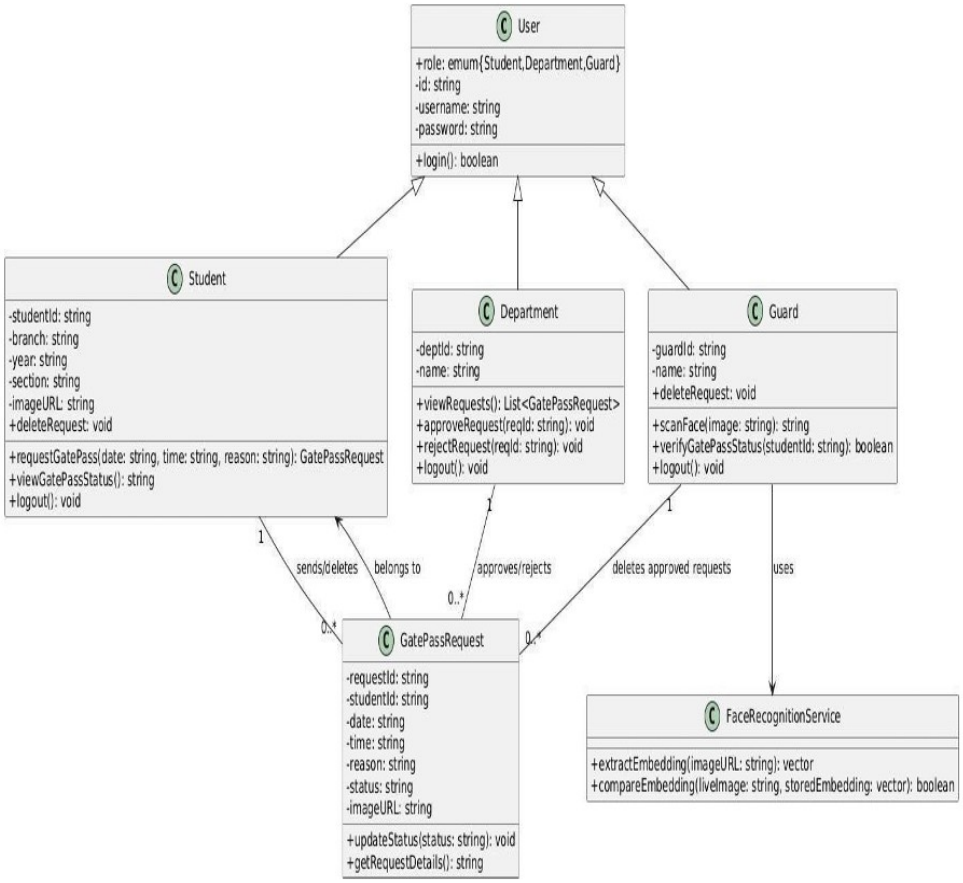


Fig 3.2.2 - Class Diagram

3.2.3 Sequence Diagram

This sequence diagram shows how students, departments, and guards interact with the NextGen Smart Pass System. Students submit requests, departments approve or reject them, and guards verify identity using face recognition before allowing exit. Each interaction follows a clear flow from login to logout, ensuring secure and structured gate pass processing.

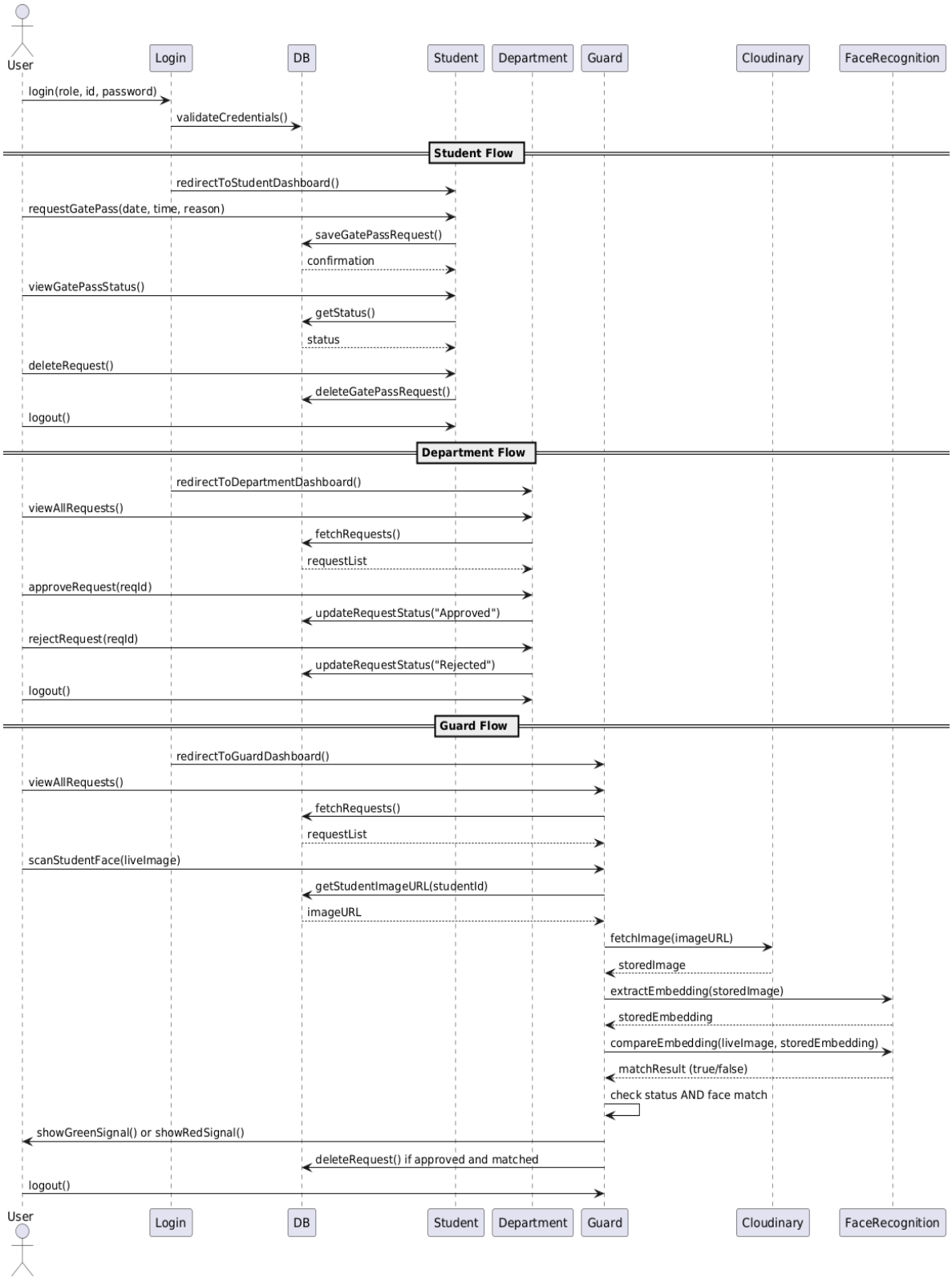


Fig 3.2.3 - Sequence Diagram

4. METHODOLOGY

4.1 Methods

The Smart Gate Pass System follows a structured digital method for requesting, approving, and verifying student gate passes.

4.1.1 Student Registration and Login

The student registers with details such as student ID, name, department, contact details, and facial image. After registration, the student logs in securely to access the gate-pass system.

4.1.2 Gate-Pass Request

The student submits a gate-pass request by entering the required details, such as date, time, and reason for leaving the campus. The request is stored in the system for further processing.

4.1.3 Parent Notification

After the student submits the gate-pass request, the system informs the registered parent about the request. The parent can view the request details and provide confirmation or rejection through the system.

4.1.4 HOD Approval

After receiving the required parent confirmation, the request is forwarded to the HOD for approval. The HOD reviews the request and can approve or reject it. The updated status is available to the student.

4.1.5 Facial Verification

When the student reaches the campus gate, the security guard captures the student's face using a camera. The system compares it with the registered facial data to verify the student's identity.

4.1.6 Gate-Pass Validation

The system checks whether the gate pass has been approved and is valid. If the student's identity matches the registered information and the gate pass is valid, access is allowed. Otherwise, access is denied.

4.1.7 Entry/Exit Record

After verification, the student's entry or exit details are recorded in the database. This provides a digital history that can be viewed by authorized users.

4.2 Tools and Technologies

The system can be implemented using:

- Frontend: HTML, CSS, JavaScript
- Backend: Python with Flask/Django
- Face Recognition: OpenCV and VGG-Face
- Database: MongoDB
- Cloud Storage: Cloudinary
- Development Environment: VS Code / Python environment
- Camera: Webcam or CCTV camera for real-time verification

4.3 Materials

The main materials required are:

- Computer or laptop
- Webcam/camera
- Internet connection
- Student and staff details
- Facial images for student identification
- Database for storing system records

4.4 Dataset

The system uses student-related data such as:

- Student ID
- Name and department
- Contact details
- Registered facial images
- Gate-pass details
- Approval status
- Exit records

The registered facial images are used as reference data for facial verification at the security gate.

4.5 System Workflow

Student Login → Gate Pass Request → HOD Approval → Approved Gate Pass → Face Verification by Security Guard → Database Matching → Access Allowed/Denied → Entry/Exit Record

The administrator manages users, gate passes, reports, and system settings. Backend services handle facial recognition, database operations, and cloud storage.

5. IMPLEMENTATION

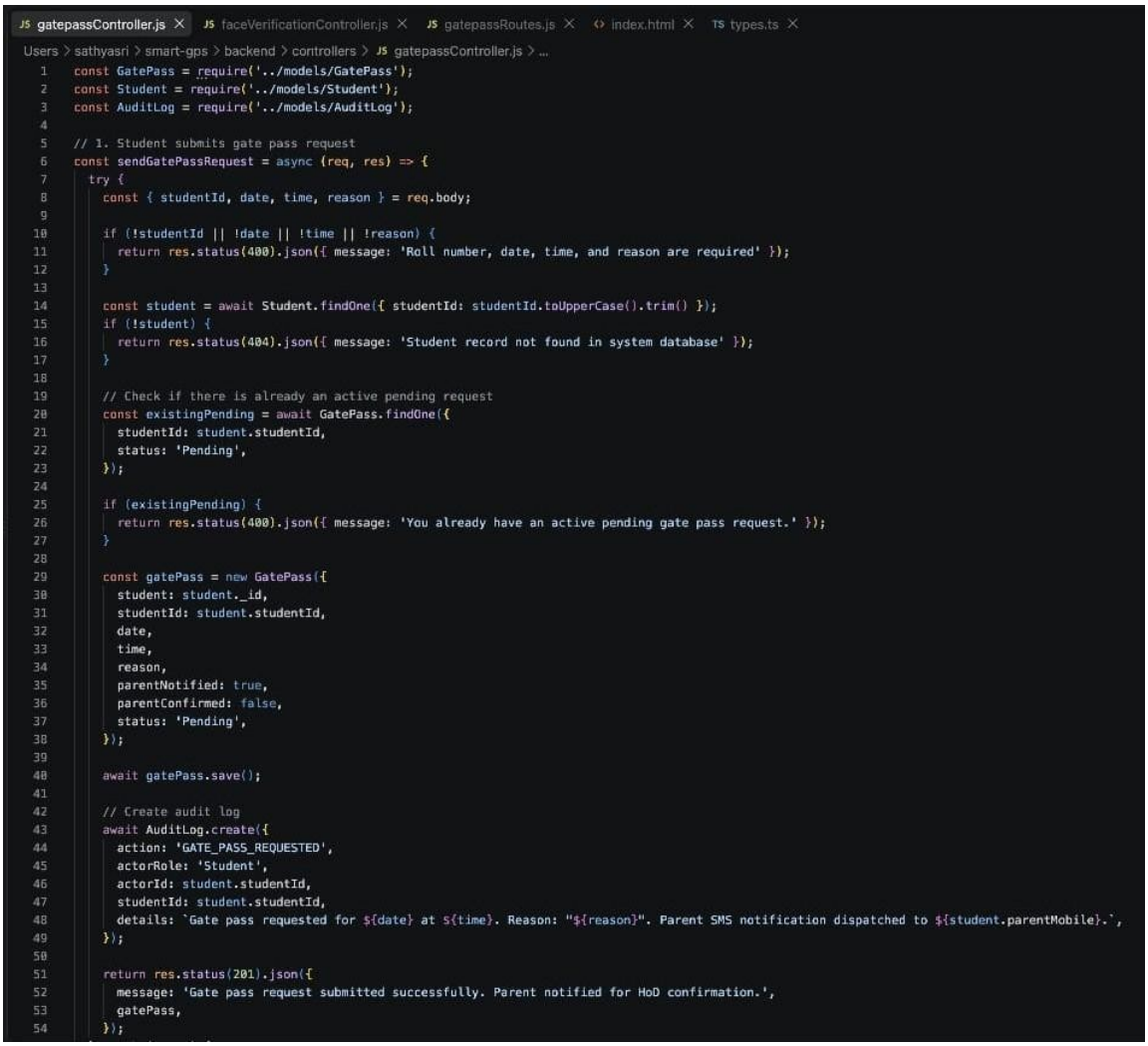


Fig 5.1 – Gate Pass Controller

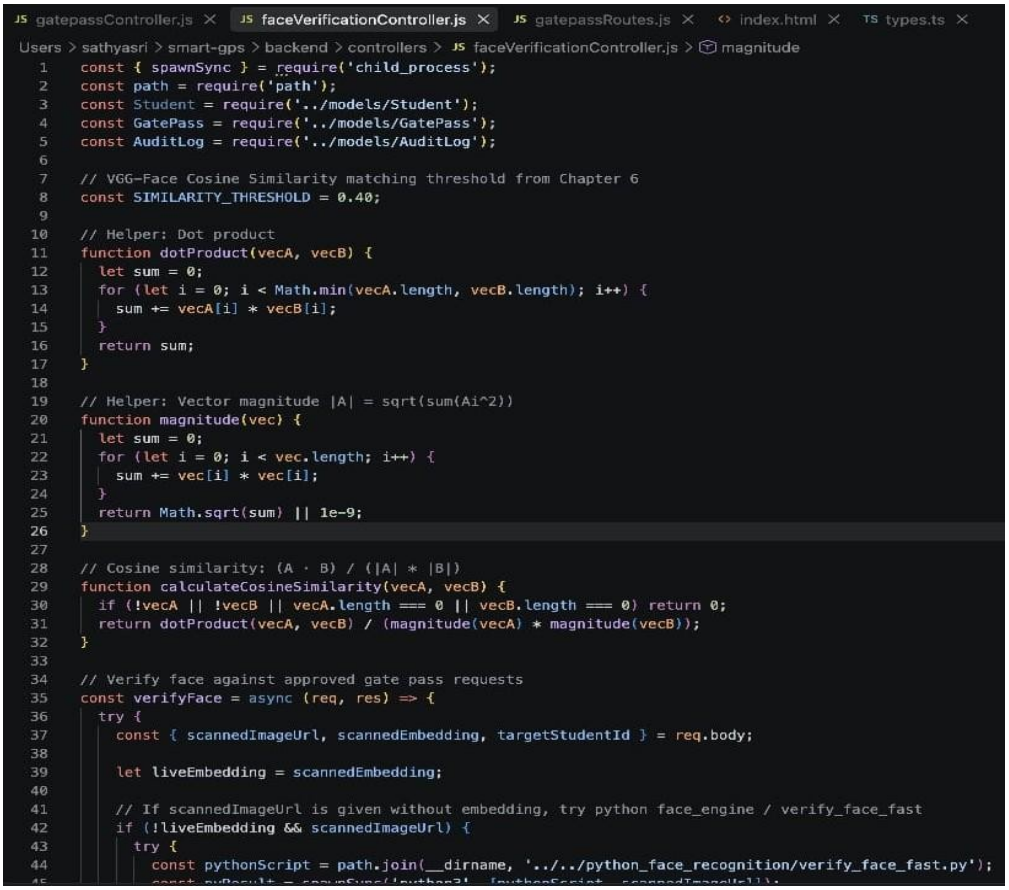


Fig 5.2 – Face Verification Controller



Fig 5.3 – Gate Pass Route

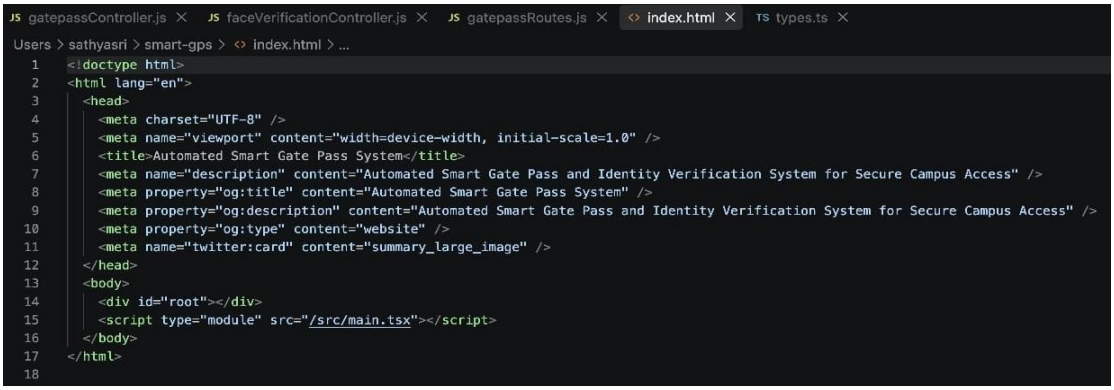


Fig 5.4 - Index

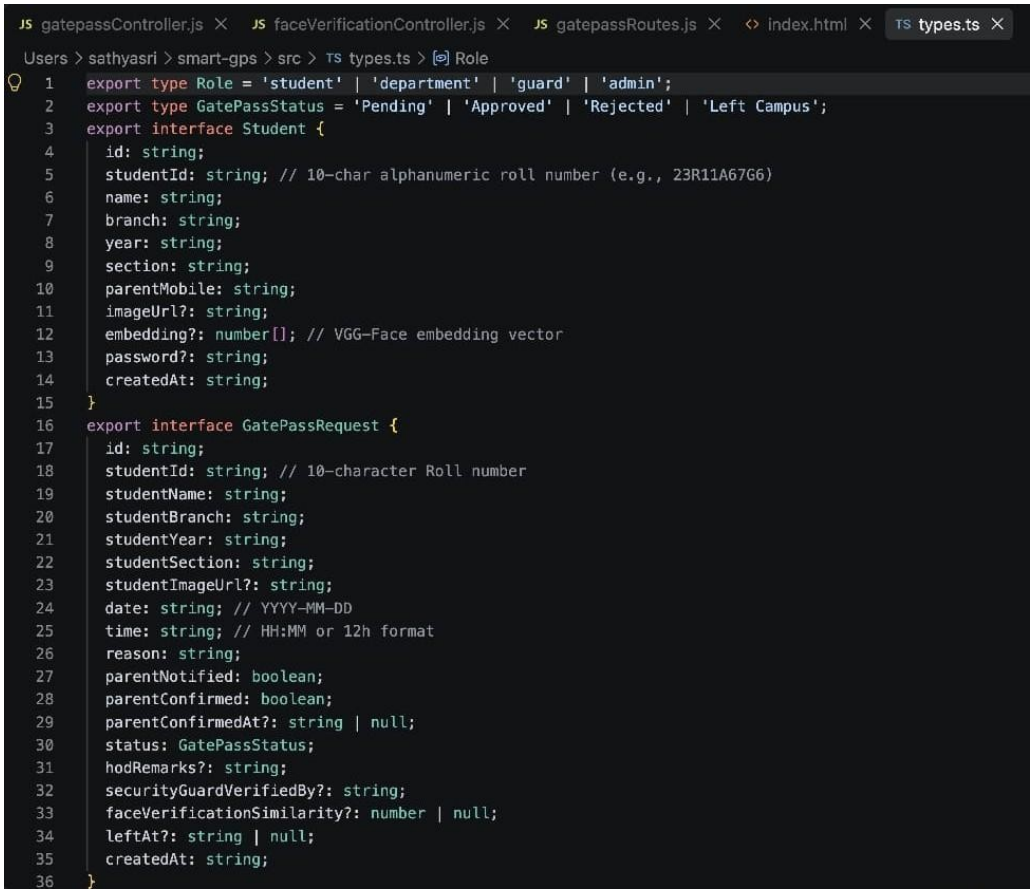


Fig 5.5 - Types

6. RESULTS & ANALYSIS

6.1 RESULTS

6.1.1 Student Registration

The student portal provides a registration form where the student enters details such as roll number, name, department, academic year, section, parent mobile number, password, and facial image. This allows the system to create a student profile and store the information required for gate-pass processing and identity verification.

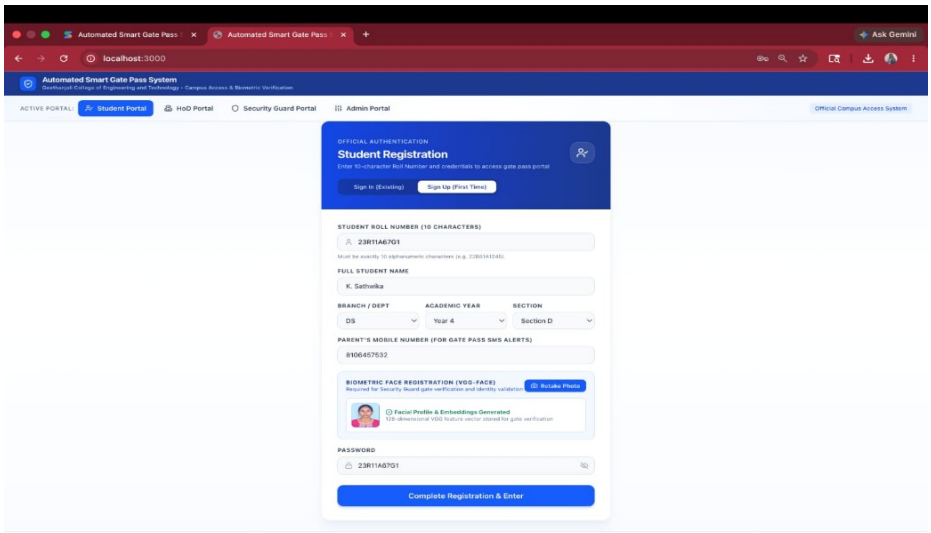


Fig 6.1.1 – Student Registration

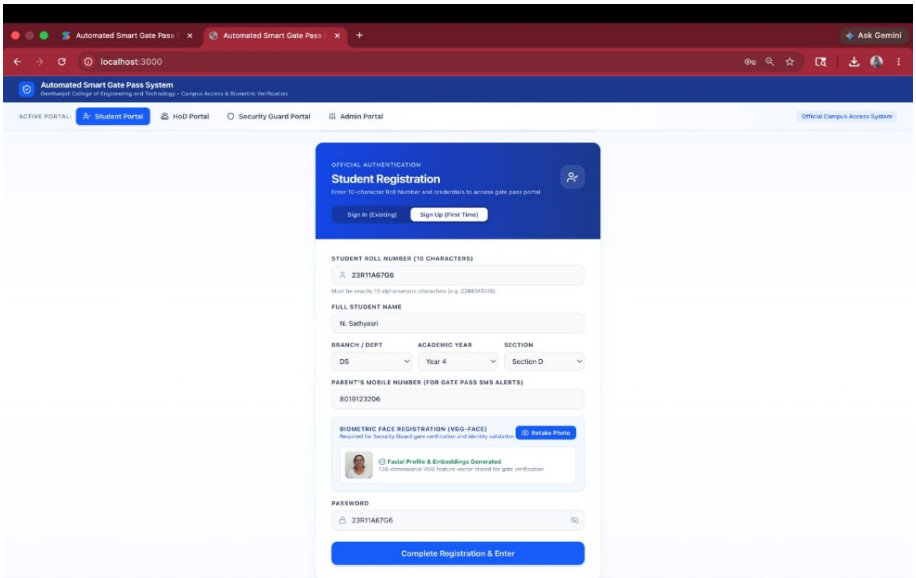


Fig 6.1.1 – Student Registration

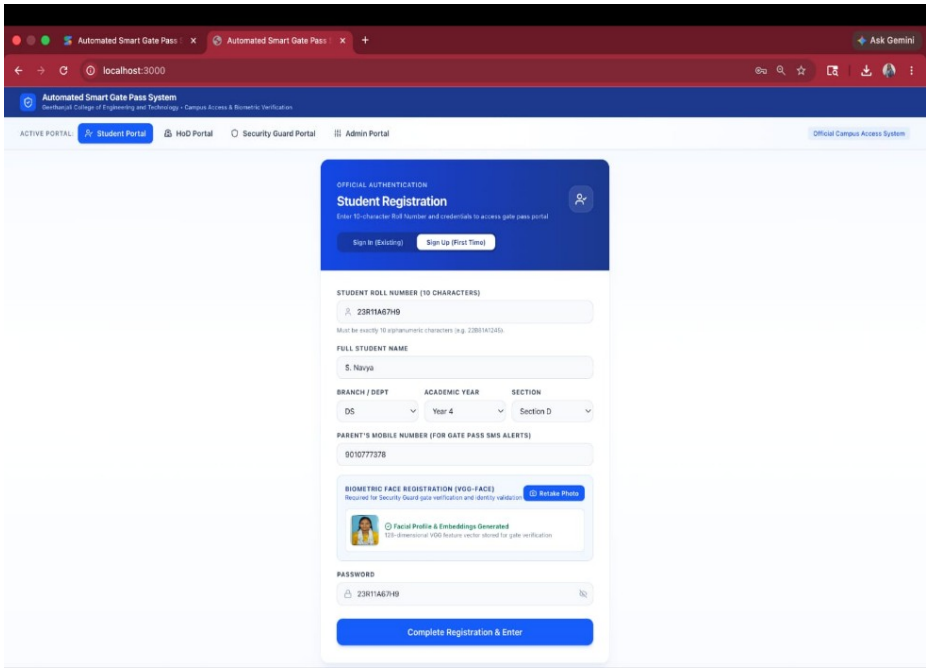


Fig 6.1.1 – Student Registration

6.1.2 HOD PORTAL

The HOD portal provides a separate login and approval interface. The HOD dashboard displays sections such as pending reviews, approved requests, rejected requests, and total requests. This provides a centralized way to review and manage student gate-pass requests.

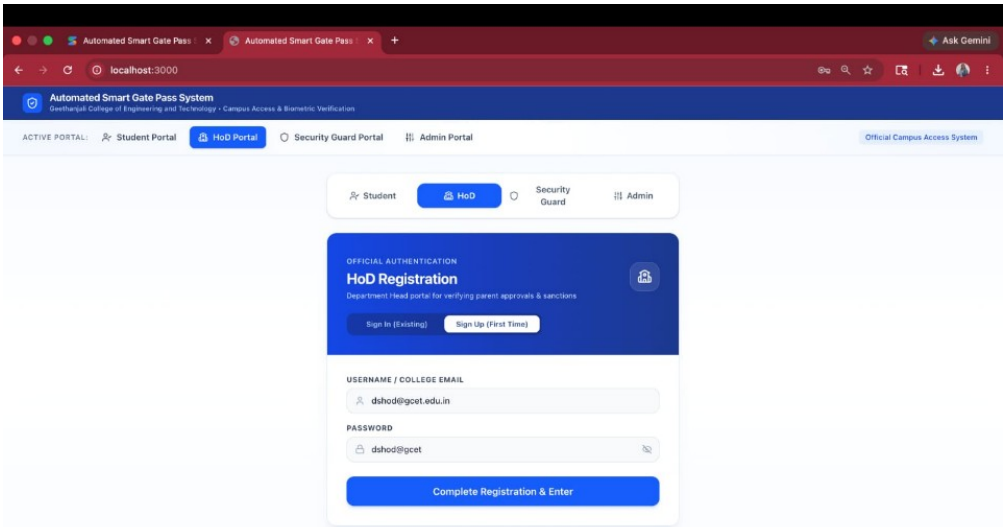


Fig 6.2 – HOD Portal

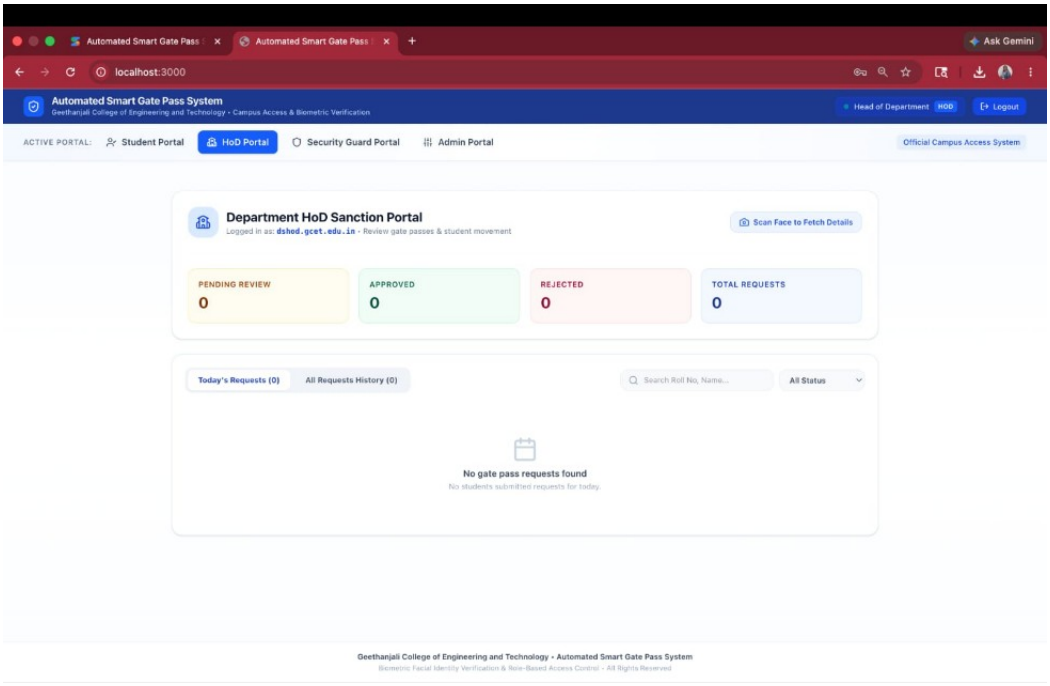


Fig 6.2 – HOD Portal

6.1.3 SECURITY GUARD PORTAL

The security guard has a separate authentication portal for gate-level verification. This role is intended to verify the student's identity and validate the gate pass before allowing campus entry or exit.

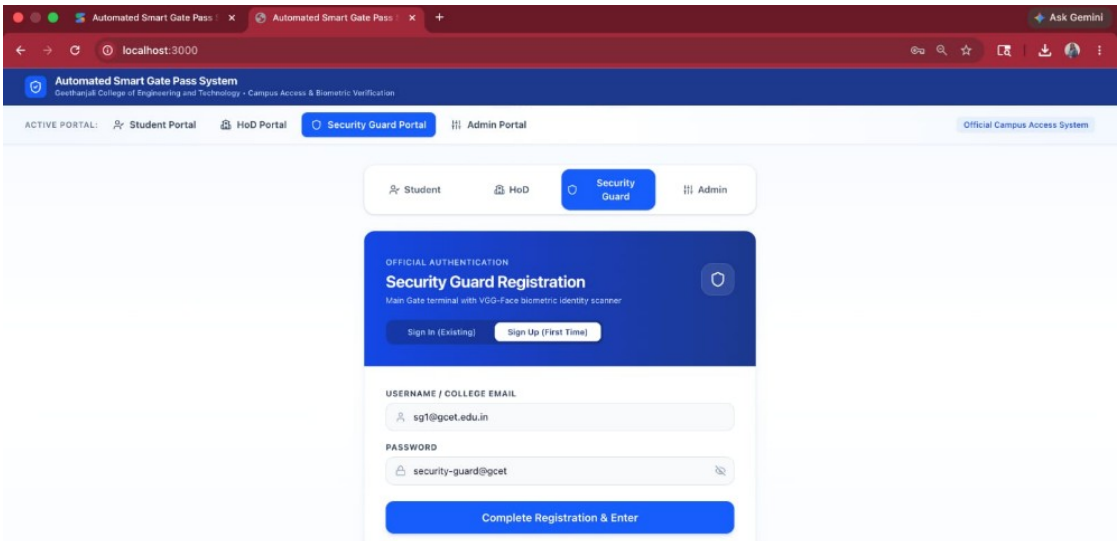


Fig 6.3 – Security Guard Portal

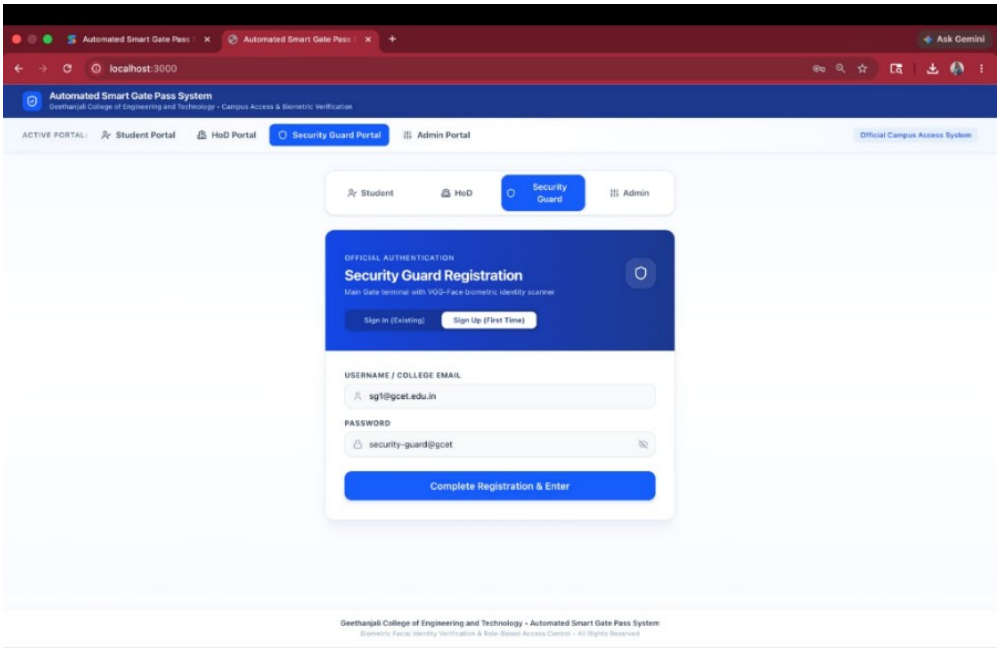


Fig 6.3 – Security Guard Portal

6.1.4 ADMINISTRATOR PORTAL

The administrator portal provides centralized system management. The screenshot shows:

Enrolled students: 3 Active passes today: 0

Campus exits recorded: 0 System audit events: 3

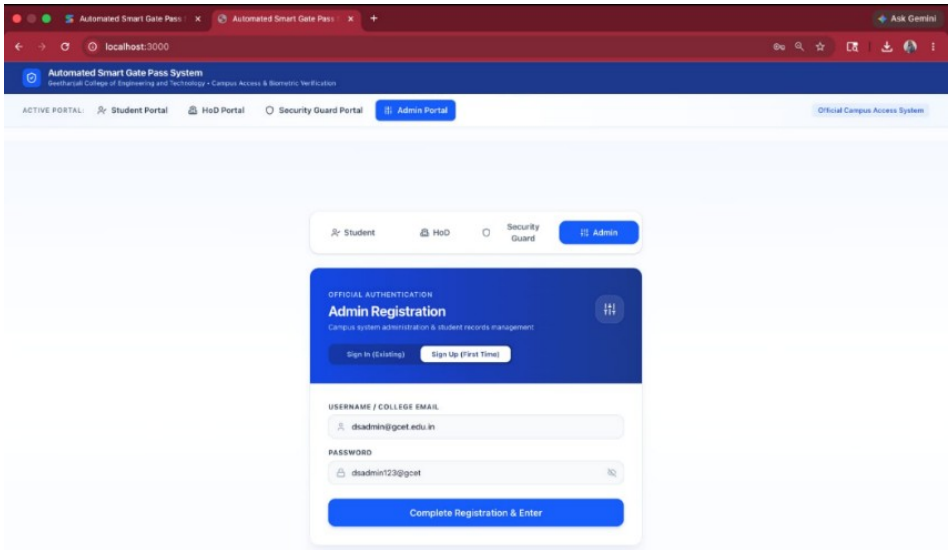


Fig 6.4 – Administrator Portal

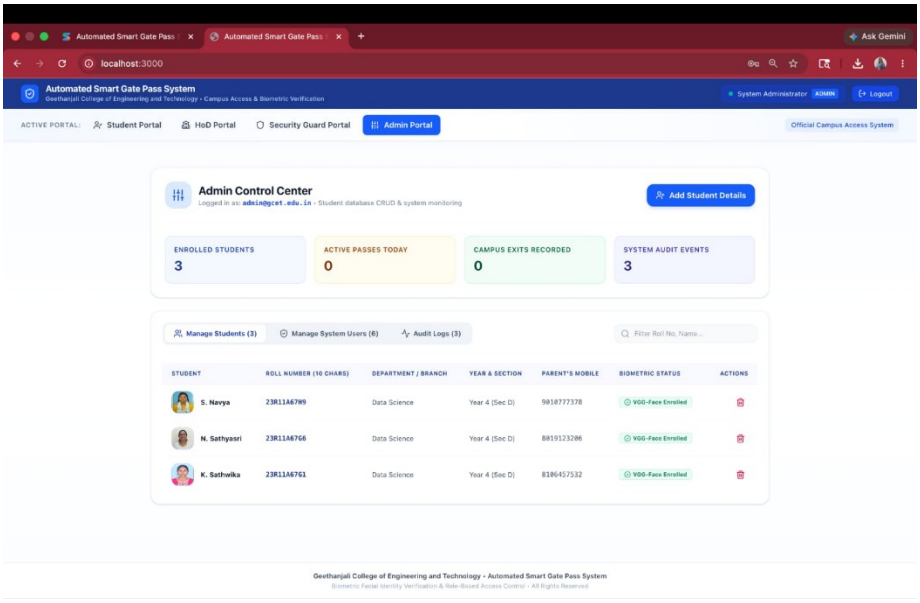


Fig 6.4 – Administrator Portal

6.2 ANALYSIS

The results demonstrate that the Smart Gate Pass System provides a structured digital platform for managing student gate passes. The system separates functions according to different user roles, including students, HODs, security guards, and administrators. Student information and parent details can be registered and maintained, while gate-pass requests can be managed through the approval process. The system also provides facilities for identity verification and maintaining entry and exit records. Overall, the implemented modules demonstrate the intended workflow of the Smart Gate Pass System and reduce the dependence on manual gate-pass management.

7. TESTING AND TEST CASES

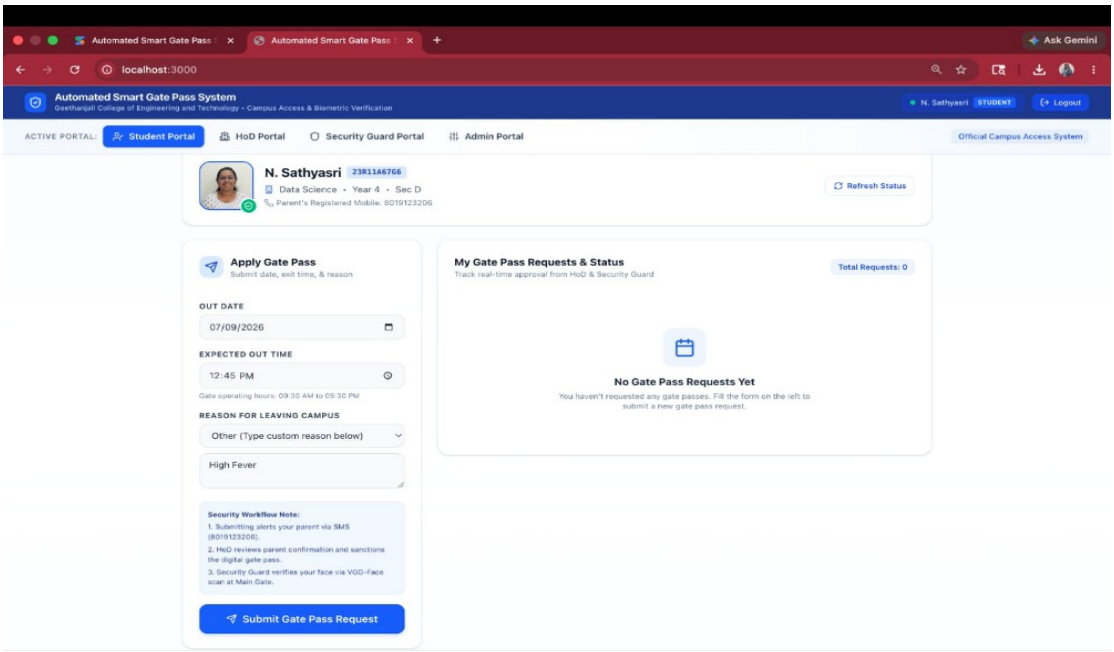


Fig 7.1 – Applying for Gate Pass Request in Student Portal

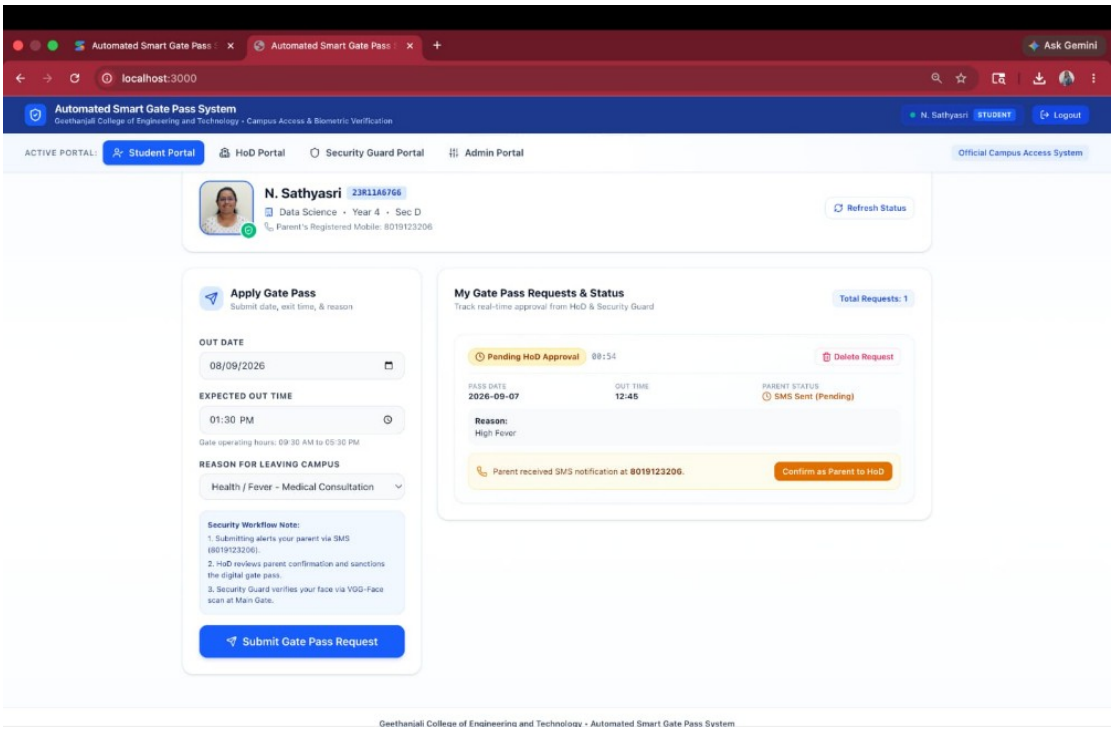


Fig 7.2 – Inform to Parent for Gate Pass Request

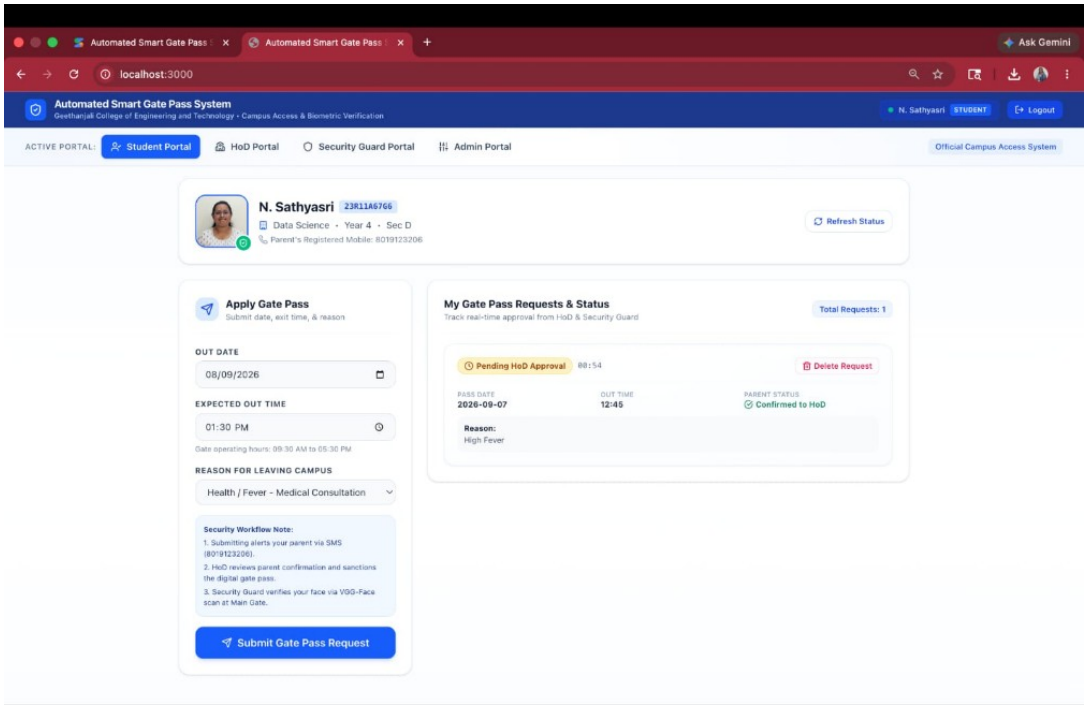


Fig 7.3 Parent's Confirmation

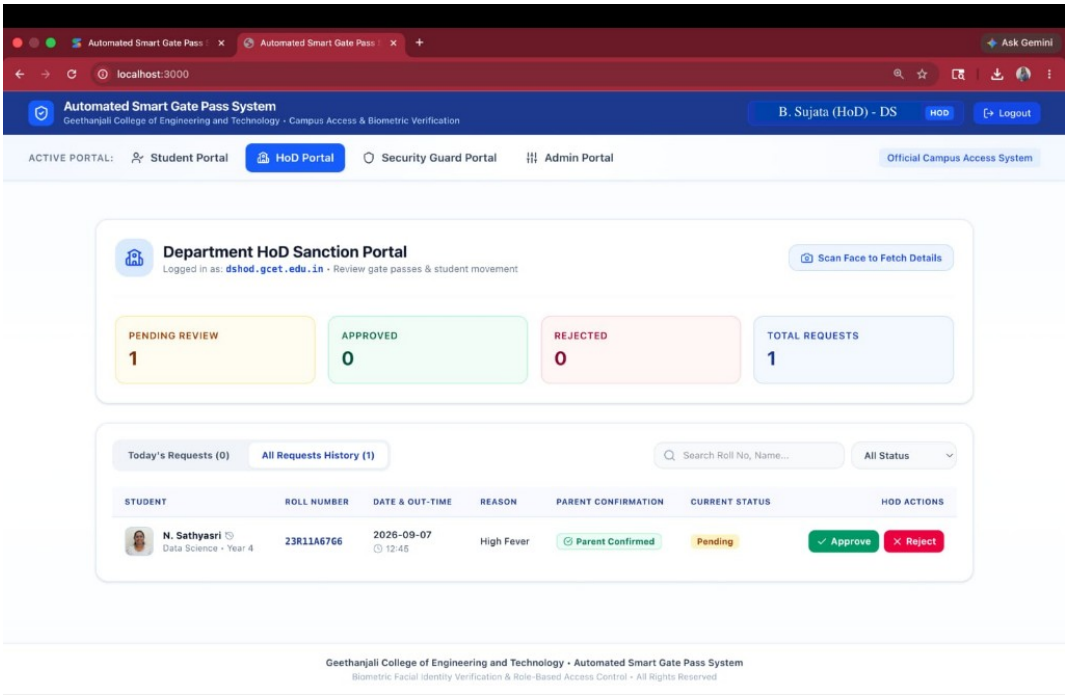


Fig 7.4 – Approve or Reject from HOD for Gate Pass

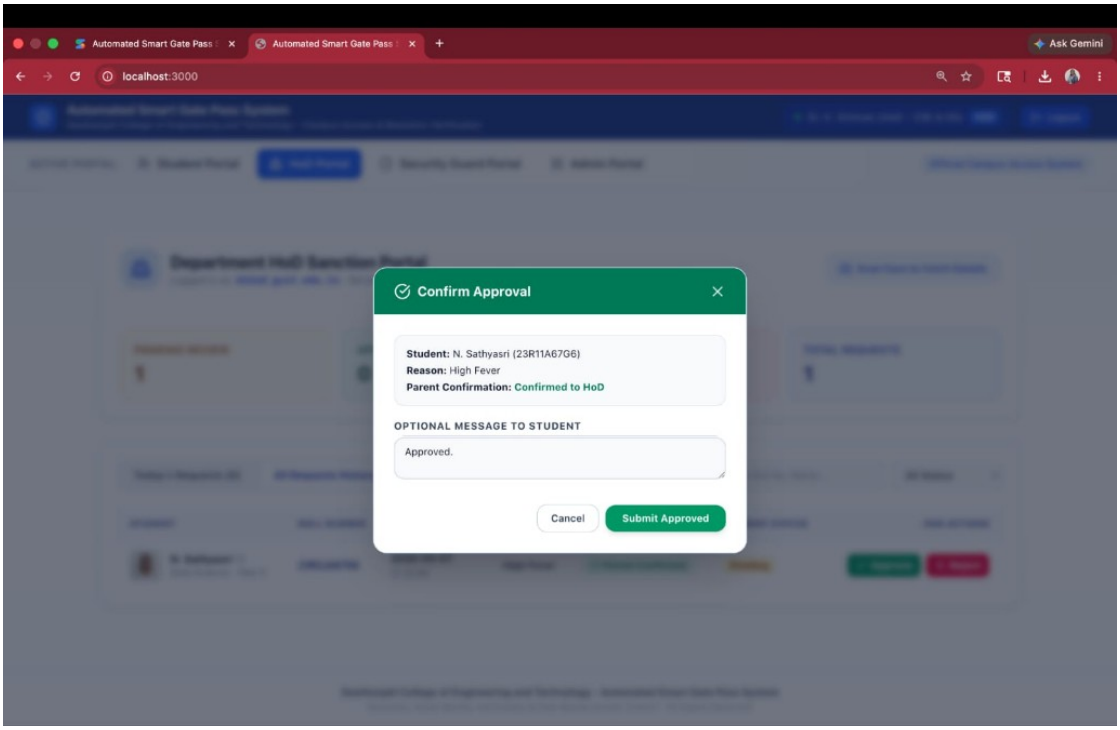


Fig 7.5 –HOD Approved for Gate Pass

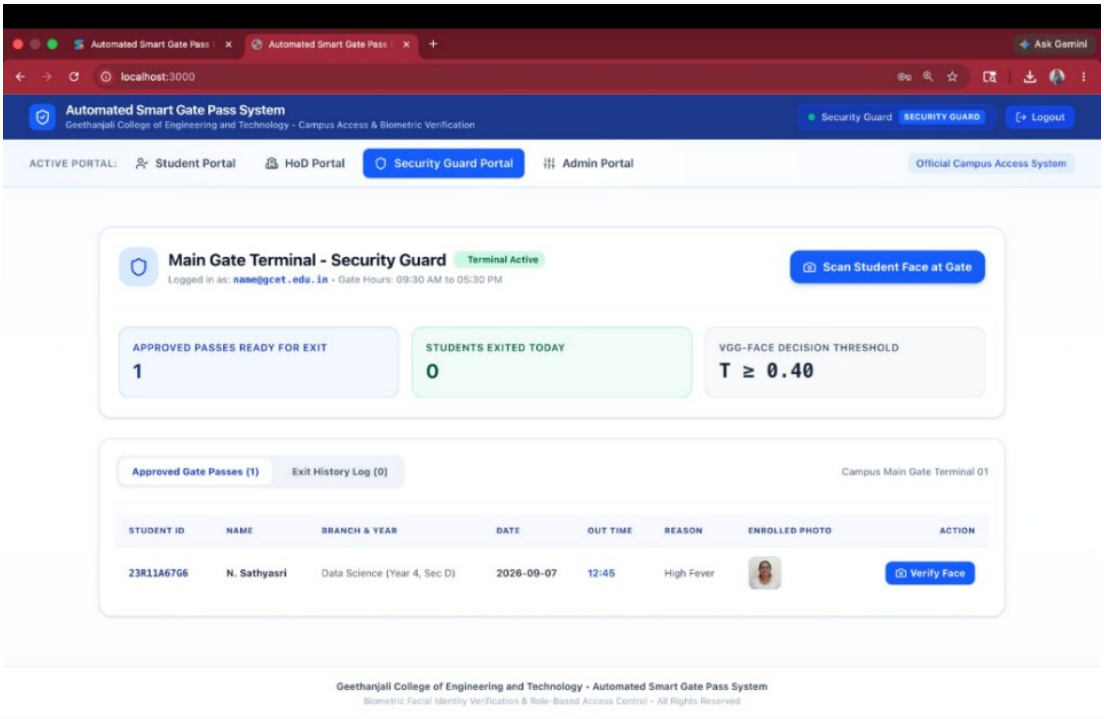


Fig 7.6 – Student at the Gate

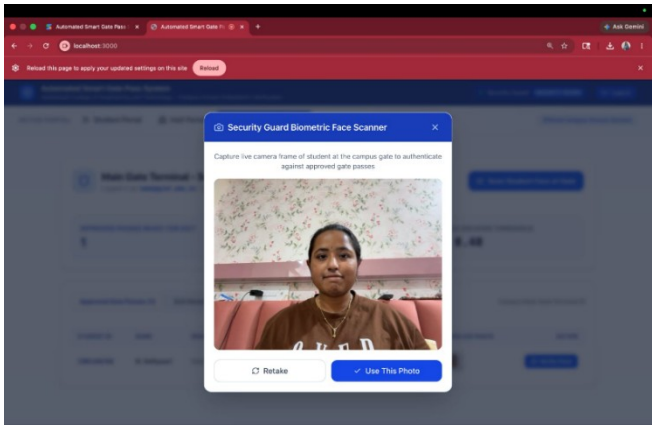


Fig 7.7 – Student Undergo for Biometric Face Scanner

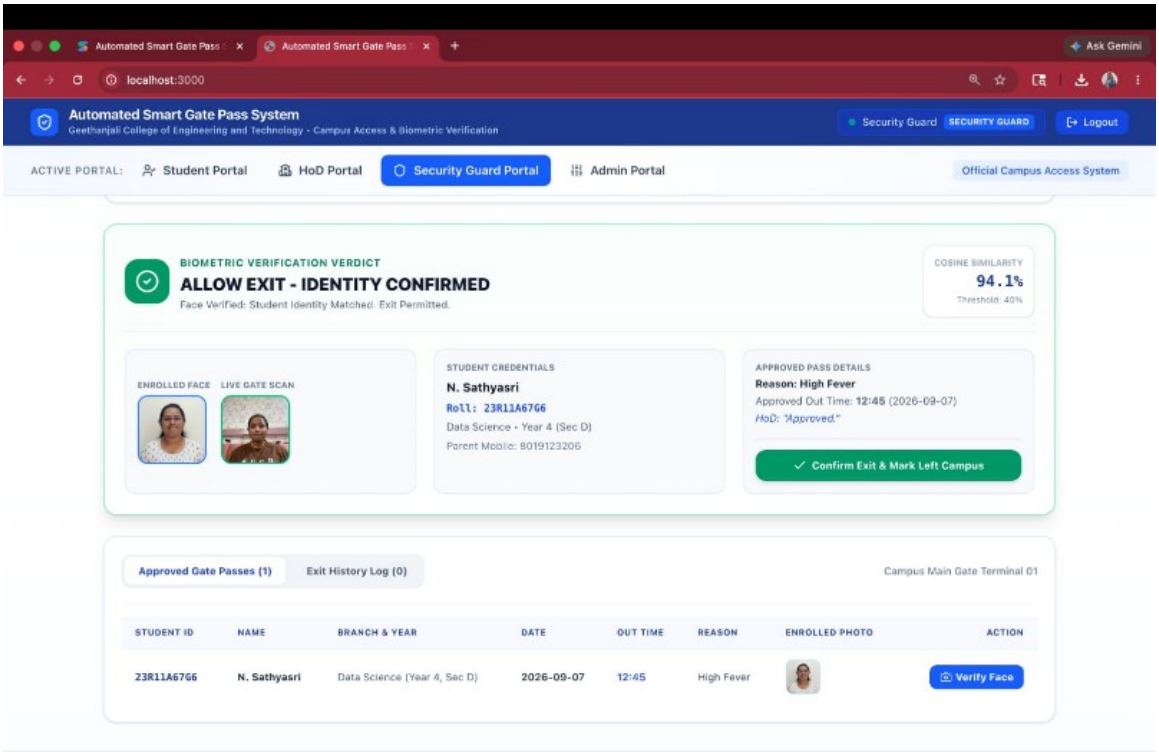


Fig 7.8 – Face Matched i.e., Allow Exit – Identity Confirmed

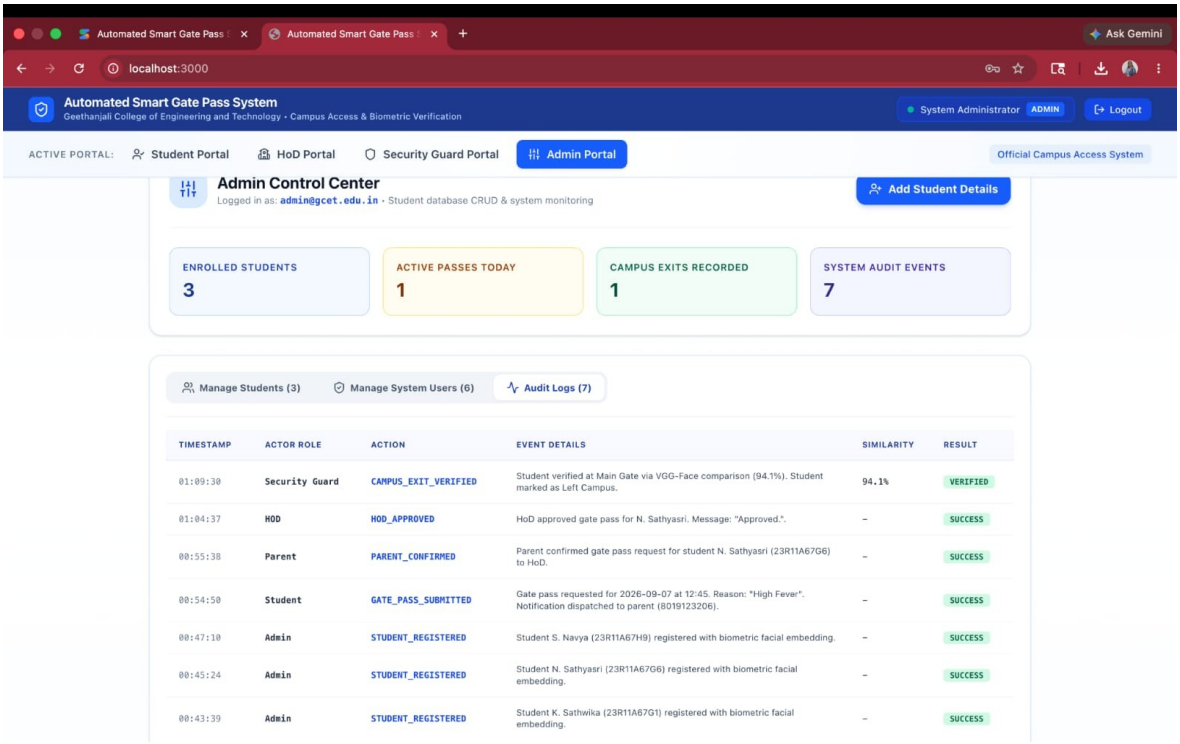


Fig 7.9 – Admin Control Center

Test Case ID	Test Case	Expected Result	Actual Result	Status
TC01	Student Registration	Student details should be registered	Details registered successfully	Pass
TC02	Student Login	Student should log in successfully	Login successful	Pass
TC03	Gate-Pass Request	Request should be submitted	Request submitted successfully	Pass
TC04	Parent Confirmation	Parent should confirm/reject the request	Confirmation recorded	Pass
TC05	HOD Approval	HOD should approve/reject the request	Request status updated	Pass
TC06	Face Verification	Registered student should be verified	Student verified successfully	Pass
TC07	Gate Access	Valid pass should allow access	Access allowed	Pass
TC08	Exit Record	Movement should be recorded	Movement should be recorded	Pass

Table No. 2 – Testing & Test Cases

8. CONCLUSION & FUTURE WORK

8.1 Conclusion

The Smart Gate Pass System was developed to provide a secure, organized, and digital method for managing student gate passes in an educational institution. The system replaces the traditional manual gate-pass process with an online workflow that makes requesting, approving, and verifying gate passes easier and more systematic.

The system provides separate functionalities for students, parents, HODs, security guards, and administrators. Students can register their details and submit gate-pass requests, while parents can be informed about the requests and provide confirmation. The HOD can review and approve or reject the requests, ensuring that only authorized gate passes proceed to the verification stage.

At the campus gate, facial verification is used to confirm the identity of the student before allowing access. The system also maintains student information, gate-pass details, approval status, and entry/exit records in a centralized database. This helps reduce manual paperwork and provides an organized record that can be accessed by authorized users when required.

Overall, the Smart Gate Pass System provides a practical approach to managing student movement by combining digital gate-pass management, approval, facial verification, and record maintenance. The implemented system demonstrates how these technologies can be integrated to create a systematic and convenient solution for educational institutions.

8.2 Future Work

The system can be further improved by adding the following features:

- Mobile application for students, parents, HODs, and security guards.
- SMS, email, or push notifications for gate-pass updates.
- Improved facial-recognition accuracy under different lighting conditions.
- QR code or digital gate-pass integration as an additional verification method.
- Real-time dashboards and detailed reports for administrators.
- Cloud-based deployment for easier access and scalability.

9. REFERENCES

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10. APPENDICES

Appendix	Description
Appendix A	User Registration and Authentication Details
Appendix B	Exit Gate Pass Request Details
Appendix C	Gate Pass Approval and Rejection Details
Appendix D	Gate Pass Status and Record Management
Appendix E	Database Structure and Sample Records
Appendix F	Test Cases and Validation Results

Table No. 3 – Appendices

CO-PO MAPPING

Course Code	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
20CS41005.1	-	-	1	-	-	-	-	2	3	-	-	-	-	-
20CS41005.2	-	-	-	-	-	-	-	3	3	-	-	-	-	-
20CS41005.3	3	3	3	-	-	-	-	3	-	-	3	3	-	-
20CS41005.4	3	-	1	-	-	3	3	-	-	-	-	-	-	-
20CS41005.5	-	-	-	-	-	-	3	-	-	3	-	-	-	-
20CS41005.6	3	-	1	-	3	-	-	-	2	3	1	-	1	-
AVG	**3.0**	**3.0**	**1.5**	**-**	**3.0**	**3.0**	**3.0**	**2.6**	**2.6**	**3.0**	**2.0**	**3.0**	**1.0**	**-**

Table No. 4 – CO-PO MAPPING